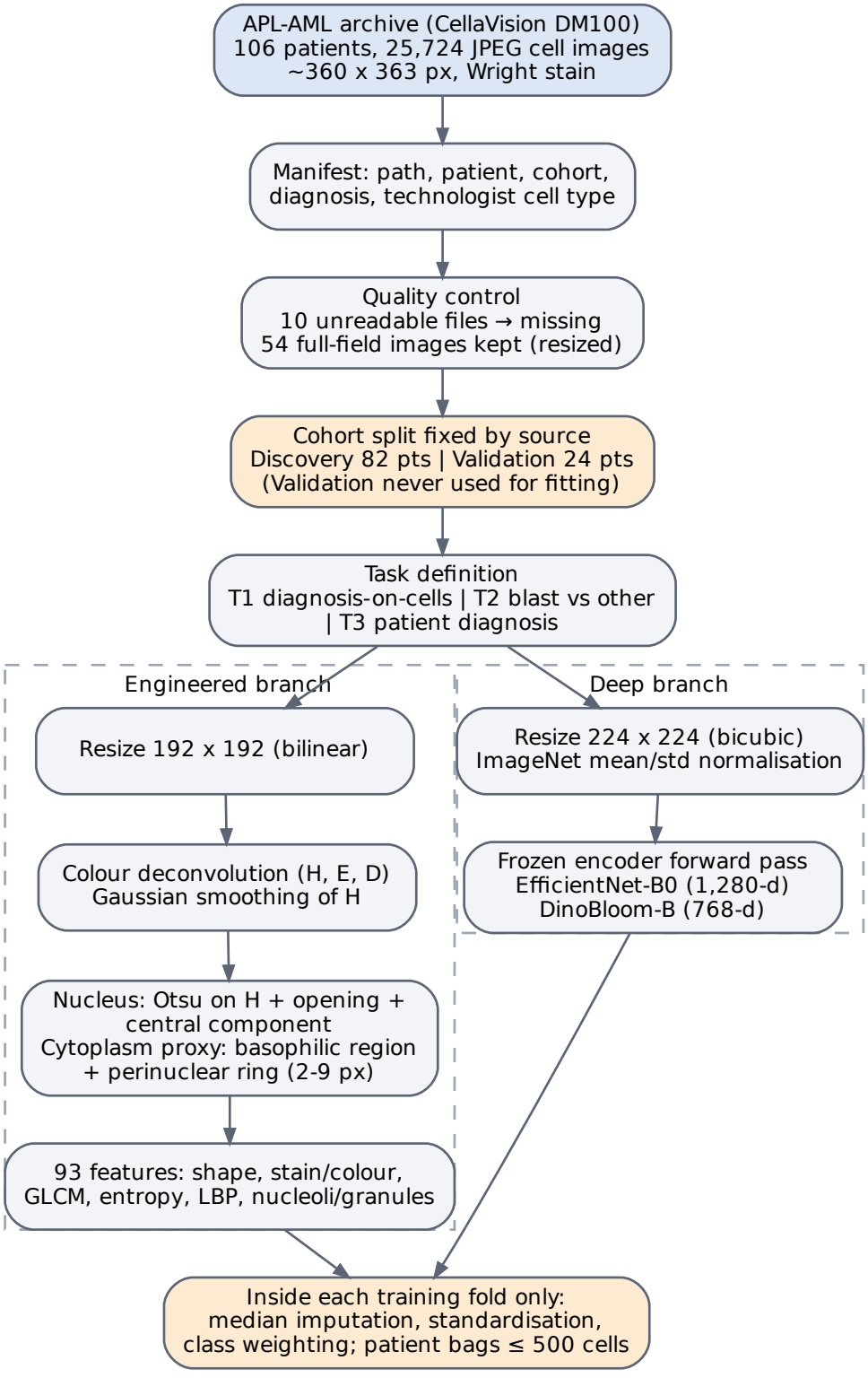
APL-AML archive (CellaVision DM100)
106 patients, 25,724 JPEG cell images
~360 x 363 px, Wright stain
Manifest: path, patient, cohort, diagnosis, technologist cell type
Quality control
10 unreadable files → missing
54 full-field images kept (resized)
Cohort split fixed by source
Discovery 82 pts | Validation 24 pts
(Validation never used for fitting)
Task definition
T1 diagnosis-on-cells | T2 blast vs other | T3 patient diagnosis
Engineered branch
Resize 192 x 192 (bilinear)
Colour deconvolution (H, E, D)
Gaussian smoothing of H
Nucleus: Otsu on H + opening + central component
Cytoplasm proxy: basophilic region + perinuclear ring (2-9 px)
93 features: shape, stain/colour, GLCM, entropy, LBP, nucleoli/granules
Deep branch
Resize 224 x 224 (bicubic)
ImageNet mean/std normalisation
Frozen encoder forward pass
EfficientNet-B0 (1,280-d)
DinoBloom-B (768-d)
Inside each training fold only:
median imputation, standardisation, class weighting; patient bags ≤ 500 cells